



Automated solvent recovery

Sustainable practices using Heidolph Distimatic rotary evaporator

The COVID-19 crisis and the resulting increased demand for disinfectants such as isopropanol or ethanol have already led to bottlenecks in some regions and to a rise in solvents prices on the world market. Therefore, solvent recycling is not only a sustainable but also a considerable way of reducing operating costs in the own laboratory and at the same time increasing efficiency.

Solvent recovery by means of an evaporation process is a common laboratory work process, for which often a rotary evaporator is used. Even if this is a suitable and sample-conserving lab device for this process, the manual operation (the so-called batch operation) is less than efficient if larger solvent volumes must be purified.

Evaporating flasks only hold a limited volume and must be refilled or replaced, which leads to long heating times of the cold medium before the process starts again. The capacity of receiving flasks is also limited and requires regular emptying during operation. To do this, the entire system must be ventilated, and a running process must be stopped.

Evaporation in batch operation is a dynamic process during which the conditions in the entire system are constantly changing.

In addition, the vacuum must be manually readjusted several times to achieve consistently high performance. Consequently, a laboratory employee is often busy for several minutes adjusting the process parameters, filling and emptying the flasks or changing them if necessary. This is a cost factor that reduces the possible savings. Hundreds of customers worldwide have already achieved a significant efficiency increase in solvent recycling by using an automated module that independently controls the filling and emptying of the evaporator and receiving flasks.

At the labs of the Faculty of Macromolecular Chemistry, University of Bayreuth (Germany), very small quantities with similar contents were very often evaporated, as usual in chemistry laboratories. The doctoral students and laboratory assistants working in this laboratory spent a significant amount of time performing the typical work steps on the rotary evaporator to purify small amounts of a few hundred millilitres of solvents.

Mr. Lothar Benker, responsible for the equipment pool in this work group, recognized the potential of automation and instructed the doctoral students in the respective work group to collect the small quantities into 5 to 20L containers and not to bother with manual processing. A Heidolph Distimatic system with a rotary evaporator automates overnight processing of a filled container after workday activation, saving hours of precious time.